

NYAKALLO PEETE

CONTACT

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EDUCATION

Mechatronics Engineering

University of Cape Town

2022 - Present

- Dean's Merit List
- Facilitator affiliated with RS Components, volunteering to guide students from disadvantaged high schools in an effort to expose them to STEM fields.
- Member of the Golden Key International Honour Society
- Class Medal: PHY1014F and PHY1015F (physics courses)
- Class Representative: MAM1024S and MAM2085F (mathematics courses)

Matric (IEB)

St Stithians College

2017 - 2021

- Academic honours in grade 11 and 12 (aggregate of above 80%)
- Academic Socratic Club (invitational club for the top 10 academic achievers in grade 11)
- Pass with distinction (2017-2021)
- Dux Scholar (2018)

HOBBIES AND INTERESTS

Stringed Instruments

Technology and Mechanics

Sports and Wellness

Arts and Crafts

PROFILE

I am a final year Mechatronics Engineering student at the University of Cape Town with a deep passion for automation, control systems and automotive technology. I have a strong commitment to collaborative teamwork in pursuit of shared goals. I am dedicated to both personal and collective growth, and am always seeking opportunities to use my skills to serve others.

WORK EXPERIENCE

Co-Founder & Systems Integration Lead

BioHydra Technologies

February 2026 - Present

Lead subsystem integration across mechanical actuation, power electronics, and embedded logic. Design all maintenance procedures and standardise communication interfaces and direct hardware-in-the-loop (HIL) testing during commissioning.

Vacation Student Intern

Exxaro Resources

Dec/Jan Holidays (2023- 2025)

2023: Engaged with Leeupan's Plant Engineering department which i comprised of Electrical Engineering, Structural Engineering, Mechanical Engineering and Instrumentation. Spearheaded a project to share a high level explanation of the pneumatic airflow across all 4 coal washing plants at the business unit to gain exposure. .

2024: Collaborated with fellow Belfast intern students to design an automated solution for their water bowser filling station . Entire vacation work exposure was with the Instrumentation department where I assisted in fault finding and component replacements. I was tasked with designing a control system for the Pit Dewatering Project. This experience provided valuable insights into leadership, communication, and teamwork, developed through collaboration with seasoned professionals.

2025: Gained exposure with Grootegeluk's Projects Department which is comprised of Electrical, Mechanical and Civil Engineering Disciplines. Gained exposure regarding general pit dewatering and was tasked with presenting the impact that an automated pit dewatering system would have on the company. The presentation focused on the predicted reduction of Scope 2 emissions, a high-level state description of the proposed system and recommendations for the current dewatering system.

Residence Admin

University of Cape Town

2023-2025

2023 - 2024: Assisted in the administration and operation of Tugwell Hall's front desk as a valued member of the Residence Operations Team. Delivered high-quality service by executing assigned duties with diligence. Public service and problem solving skills were developed during my term.

2025: Promoted as Subwarden. Worked closely with 5 peer-colleagues and the residence Warden and Deputy Warden in discussing residence matters, enforcing residence and university rules and offering first-point-of-call assistance for residents whilst day-staff members are off work. Portfolio work included: overseeing the reception with the Reception Coordinator, organising weekly Residence Management Team meetings, coordinating HR and application processes.

FSA UCT ELECTRICAL LEAD

University of Cape Town

2024-2025

Distinguished leader of UCT's esteemed Formula Student Africa (FSA) Electrical Team which is participating in a rigorous competition focused on designing and testing an electrical Formula 1 car. My responsibilities entailed overseeing 6 sub-team leads and 28 team members, setting up and chairing bi-weekly meetings and overseeing the overall research conducted by the electrical team.

REFEREES

Dr David Oyedokun - Academic Advisor

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Mr Dominic de Maar - Chief Tech. Off.

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SKILLS

- Crisis Management
- Detail Oriented
- Troubleshooting
- Adaptability
- Team Coordination
- Time Management